A Shift Reduction Domain Generalization Network for Hyperspectral Image Cross-Domain Classification

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***Abstract*—In practical applications, varying imaging conditions cause spectral shifts for the same class across hyperspectral image (HSI) domains. Besides, since annotating HSIs is time-consuming, the number of available labeled samples is insufficient, resulting in strong models to overfit during training. To address these issues, a spectral–spatial structure refactoring-based data representation method and a lightweight shift reduction domain generalization network (SRDGnet) are proposed for cross-domain classification. Specifically, a spectral-spatial exchange attribute extraction module (SE-AEM) is designed to capture fine-grained spectral local features, while an integrated feature extraction head (IFEH) is used to extract global features. They both map their respective features into a low-dimensional space. Subsequently, a domain shift reduction module (DSRM) fuses the features for information interaction to reduce domain shift and then feeds them to a lightweight version of the integrated feature extraction module (IFEM)-light for domain-invariant and discriminative features of classes, which are then used for final classification. Moreover, the training leverages multisource domains and a batch-wise traversal constraint strategy to enhance sample diversity and utilization. Experimental results on four public HSI datasets demonstrate that the proposed method can effectively learn a model with higher generalization ability and stability. On the Houston 2013 dataset, it achieves an overall accuracy (OA) of 89.35%, which is 2.67% higher than the best performance of other comparison methods. The code will be released soon on the site of https://github.com/Cherrieqi/SRDGnet**

***Index Terms*—Cross-domain classification, domain generalization (DG), hyperspectral images (HSIs), multisource domain.**

I. Introduction

HYPERSPECTRAL images (HSIs) offer the advantages of high spectral resolution and integrated spectral-spatial information, enabling their unique applications in many fields [1]. HSI classification technology allows for fine-grained classification of ground objects and plays a significant role in mineral resource exploration, military camouflage recognition, precise vegetation differentiation, urban system management, and so on, attracting widespread attention from researchers [2].

The high dimensionality of the spectral dimension in HSIs results in sparse data distribution and challenges in feature modeling [3]. Deep learning excels at automatically extracting high-level nonlinear features from images, eliminating the step of manual feature extraction [4]. Consequently, numerous deep learning-based methods for HSI classification have emerged, employing various types of neural networks to mine spectral-spatial features [5], [6], [7], [8], [9]. These methods have demonstrated significantly superior classification performance compared to traditional machine learning approaches.

With advancements in imaging technology and growing application demands, high-precision HSI classification techniques increasingly rely on deep learning [10], which requires sufficient labeled data for model optimization. However, limited spatial resolution makes manual annotation time-consuming and costly, and newly acquired HSIs often lack sufficient labels. Consequently, many researchers have shifted toward few-shot HSI classification [11]. An early fundamental approach involves methods based on in-domain sample sets, aiming to extract as much useful information as possible from the available data. Common strategies include designing improved feature extraction techniques and employing more effective data augmentation methods to enrich the training set [12], [13], [14], [15], [16]. Another line of research focuses on expanding the in-domain sample set by generating pseudo-labels. This typically involves selecting a portion of unlabeled data from the current operational domain and assigning pseudo-labels, thereby increasing the size of the training dataset [17], [18], [19], [20], [21].

Due to limited available data, the performance gains of these methods are inherently constrained. Therefore, a cross-domain transfer learning strategy starts to be adopted: leveraging the knowledge of ground object classes from label-rich domains to assist classification in the target domain (TD). It has given rise to the task of HSI cross-domain classification. Most methods in this field adopt a deep learning-based domain adaptation (DA) framework, which can be broadly categorized into discrepancy-based methods and adversarial learning-based methods [22]. Discrepancy-based deep DA methods aim to align the marginal and/or conditional distributions between domains by introducing adaptation layers into deep neural

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networks for specific task representations [23], [24], [25], [26], [27]. In contrast, adversarial learning-based methods learn transferable and domain-invariant features through adversarial learning [28], [29], [30], [31], [32], [33], [34].

DA methods require image samples from the source domain (SD), labels for SD samples, and image samples from the TD [35]. Labels for TD samples are not directly used to supervise the training of models. However, aligning the class distributions between the two domains implicitly involves the use of TD labels. Consequently, these methods face significant limitations in practical applications. To better meet application demands, a method that entirely avoids relying on TD labels is needed. In recent years, domain generalization (DG) has garnered increasing attention and has been widely applied in various fields. DG aims to learn a highly generalized model from one or more different but related SDs so that the learned knowledge can be effectively transferred to a TD. Importantly, DG does not require access to TD samples during training. Over the past few years, DG has made significant progress in the field of computer vision. A style normalization and modulation framework has been introduced to better separate task-relevant features from task-irrelevant ones while ensuring high generalization and discriminative capabilities of the model [36], [37], [38], [39], [40], [41].

As a result of the complexity of imaging conditions and the unique characteristics of HSI data, DG methods developed for natural images are often not directly applicable to HSIs. Currently, research on HSI cross-domain classification using DG is less. Most existing studies focus on single-source DG, where extended domains (EDs) are constructed by generating samples and feature-level comparisons are made with the SD to achieve generalization [42], [43], [44], [45], [46], [47]. Nevertheless, the performance of single-source DG models often depends on the similarity between the SD and TD, resulting in insufficient generalization capability. Therefore, we previously proposed a multisource DG method for HSI cross-domain classification to enhance model's generalization ability [48]. However, the method has a relatively large number of network parameters and high model complexity, which requires more labeled samples during training to reduce overfitting. Consequently, when the number of labeled samples is insufficient, the model's classification performance may fluctuate significantly, leading to instability.

To overcome such limitations, a multisource DG framework for HSI called the shift reduction DG network (SRDGnet) is proposed. It aims to develop a model with strong discriminative feature extraction capabilities and low complexity for HSI cross-domain classification tasks, thereby improving the model's generalization ability under conditions of limited data amount conditions. The proposed SRDGnet introduces lightweight designs for each module. Specifically, corresponding components are improved to enhance fine-grained feature extraction. Moreover, the batch-wise constraint mechanism in model training is adopted to mitigate the drawbacks caused by limited data availability.

Our main contributions are summarized as follows.

1) To mitigate the overfitting phenomenon, a lightweight network structure is designed from the perspective of domain shift reduction. This approach enhances the utilization of features extracted by different modules and eliminates unnecessary network parameters.
2) A novel data representation method is devised, and the spatial exchange attribute extraction module (SE-AEM) is proposed to adapt to the specific data format. The purpose is to enhance fine-grained local spectral features, providing more detailed information for subsequent domain shift calculations, while reducing the number of parameters.
3) The IFEM-light is proposed to mitigate feature blurring caused by increased network depth while maintaining a lightweight module structure.
4) A batch-wise sample pair traversal approach is employed to constrain feature similarity, improving data utilization and alleviating the conflict between limited data and complex networks.

The rest of this article is organized as follows. The related work is presented in Section II. The details of the proposed SRDGnet are described in Section III. The experimental setup and results are shown in Section IV. Finally, the conclusion is given in Section V.

II. Related Work

In this section, relevant research of this article is briefly reviewed, including DG, single-source DG methods for HSI cross-domain classification, and multidomain DG methods.

A. Domain Generalization

DG presents a greater challenge than DA, as DG aims to train a model using only the SD data, without access to TD data during the training phase. The trained model is then expected to generalize effectively to the TD during inference. Existing DG approaches are generally classified into three main categories: data manipulation, representation learning, and learning strategy [35].

1) Data Manipulation: This category of methods emphasizes modifying inputs to support the learning of general representations. Within this approach, two popular techniques are widely used. One is data augmentation, which primarily involves augmenting, randomizing, and transforming input data [49], [50]. The other one is data generation, which creates diverse samples to enhance generalization [36], [51].

2) Representation Learning: This category of methods is the most widely used in DG. The first key technique is domain-invariant representation learning, which employs kernel methods, adversarial training, explicit feature alignment across domains, or invariant risk minimization to learn representations that are invariant to the domain [37], [38]. The second is feature disentanglement, which aims to separate features into domain-shared and domain-specific components to improve generalization [40], [41].

3) Learning Strategy: This category of methods emphasizes leveraging a general learning strategy to enhance generalization capabilities. It includes ensemble learning, meta-learning, gradient operation, distributionally robust optimization, self-supervised learning, and so on [52], [53], [54], [55], and [56].

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B. Single-Source DG Methods for HSI Classification

There are a few related works on single-DG for cross-domain classification of HSIs, mainly by generating ED samples to learn domain-independent representations.

One of the earliest works is a single-SD expansion network (SDEnet) proposed by Zhang et al. [42]. A generator is designed with a semantic encoder and a morph encoder to produce an ED using an encoder-randomization-decoder structure. In addition, supervised contrastive learning is applied in the discriminator to capture class-specific domain-invariant representations, aligning intra-class samples between SD and ED. Adversarial training is further used to optimize the generator, ensuring intra-class samples from SD and ED are effectively separated. Then, Zhang et al. [43] proposed the language-aware DG network (LDGnet). It employs a dual-stream architecture, consisting of an image encoder and a text encoder, to extract both visual and linguistic features. In this setup, coarse-grained and fine-grained text representations are used to capture two levels of linguistic features. Moreover, the linguistic features serve as a shared semantic space across domains, enabling visual-linguistic alignment by supervised contrastive learning within this semantic space.

Inspired by the previous two works, Zhao et al. [44] proposed a locally linear unbiased randomization network (LLURnet). It introduces an extension generation network, featuring a symmetric encoder–decoder structure to implicitly capture local joint features under style randomization. Similarly, Wang et al. [45] propose a generalization network that leverages nonlinear sample generation. It separates sample features into invariant and variant components, generating new samples by applying nonlinear transformations to the variant features.

Although these methods have good performance in some combinations of source and TD datasets, the generalization performance of single-source DG is far from enough.

C. Multisource DG Method for HSI Classification

At present, research on the multisource DG method for HSI classification remains limited. The representative work is a multisource DG two-branch network (MDGTnet) proposed by us for cross-domain classification of HSI [48]. This method uses the idea of combining disentanglement and feature alignment for DG. MDGTnet is composed of a classifier and two branches: the intradomain uniqueness extraction branch (intra-DUEB) and the interdomain commonality extraction branch (inter-DCEB). The intra-DUEB focuses on extracting internal attributes specific to each SD, which represent the characteristics of the imaging environment. The inter-DCEB, on the other hand, is designed to capture common features shared across different SDs. The features extracted by these two branches are fused at multiple levels to mitigate the impact of varying imaging environments, allowing for more discriminative features of classes.

Specifically, in intra-DUEB, the AEM is designed to segment the spectral bands for accurate spectral feature representation while also calculating spectral weights to preserve the original consistency among spectra. In addition, inter-DCEB consists of an integrated feature extraction head (IFEH) and an integrated feature extraction module (IFEM). IFEH is used to perform preliminary feature extraction, and IFEM is designed to extract multilevel features from different SDs. The model is trained using a contrastive learning approach, incorporating classification, similarity, and similarity-value difference progressive (SDP) constraints.

To sum up, the core idea of MDGTnet involves constructing intra-DUEB and inter-DCEB to calculate environmental features and class features separately. Each branch consists of two modules, with the second module progressively fusing environmental features of different levels with the corresponding class features. As a result, the two branches remain relatively independent during the feature extraction process in the first module, and interaction between them occurs only in later stages. This way relies on additional parameters to align feature levels and achieve information exchange. Training such parameters requires sufficient samples. With limited labeled HSI data, overfitting is likely to arise.

III. Proposed Method

In this section, motivation, details of data structure refactor, design of the proposed SRDGnet, construction strategy of sample pairs, and loss functions are presented, respectively.

A. Motivation

With reference to the analysis on the limitations of existing methods, the proposed method is designed inspired by MDGTnet [48], while we make the considerations from the following three aspects: network structure lightweighting, fine-grained feature extraction, and efficient utilization of limited data.

1) Network Structure Lightweighting: If the internal attributes specific to each SD and the common features shared across different SDs could interact earlier and be mapped to the same low-dimensional feature space, additional modules for feature-level alignment would no longer be necessary, avoiding an increase in parameter count. Feature dimensionality reduction allows for more concise information representation. Therefore, the overall process of feature extraction is considered from the perspective of reducing domain shift, with information interaction conducted in the early stages to minimize shift, rather than separating intra-domain and inter-domain feature extraction. In addition, under the above premise, each module used in the network can also be designed in a lightweight manner.

2) Fine-Grained Feature Extraction: HSIs have a 3-D spatial–spectral structure with relatively low spatial resolution. Many land-cover classes that depend on subtle spectral differences are spatially similar or lack distinct spatial features. Two-dimensional convolution, which reduces model parameters through parameter sharing and sparse connections, acts on the spatial dimension. For HSI data input as small patches, convolution and fully connected are essentially equivalent in the spectral dimension, yielding global features but lacking detailed information. Furthermore, due to the high spectral dimensionality, a large number of convolution channels are often required to prevent information loss, contributing to a

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significant increase in model parameters. For this reason, a new data representation method needs to be innovated to make the HSI small-patch more suitable for convolution operations. The new representation can swap the spatial and spectral dimensions without altering the fundamental physical meaning of the data, to facilitate subsequent feature extraction.

3) Efficient Utilization of Limited Data: Although the number of samples is fixed, the way of constructing positive and negative sample pairs is not. Therefore, during model optimization, constraints on sample pairs can be applied batch-wise. This permits the construction of more diverse sample pairs with corresponding constraints.

B. Details of Data Structure Refactor

Owing to the correlation between adjacent spectral bands, it is essential to preserve the physical meaning of the original data structure during the spectral dimension rearrangement in a 2-D space. Therefore, the following three principles must be followed.

First, the original order of spectral band connections must be preserved, meaning that after refactor, the neighboring bands of a central band should still be those originally closer to it, while bands originally farther away should remain outside the neighborhood. Second, translational invariance must be maintained, ensuring that when the convolution kernel translates over the refactored data, the structure of each data patch within the scope of the convolution kernel remains the same, with only the band positions changing. Finally, translational continuity must be ensured, meaning that when the convolution kernel moves over the refactored data with a stride of 1, the spectral neighborhood included in the current convolution patch changes continuously compared to the previous one, thereby preventing information loss.

What is more, since HSI data patches cover only a small spatial neighborhood, and spectral features are more critical than spatial features, the spatial dimension can be flattened directly. The fully connected action of the channel dimension in the 2-D convolution kernel encompasses all spatial information, serving a similar role as before the refactor.

Specifically, for each pixel in the original HSI patch, the spectral bands are rearranged in 2-D space, and the rearranged patches are then aligned along the channel dimension. The operation process details of the data structure refactor are presented in Fig. 1. Assuming the original data patch shape is (c, h, w), where c, h, and w represent the number of spectral bands, height, and width of the HSI patch, respectively. Considering the uniformity of the arrangement of the refactored bands on the square space, the rearrangement of bands is performed by taking $(c + 2)/3$ consecutive bands horizontally and selecting bands vertically in an alternating pattern, skipping every other band (i.e., in odd-numbered order). The corresponding positions of each spectral band in the original HSI patch within the refactored data patch are shown in Fig. 1, where $B_1, B_2, ..., B_c$ represent the indices of the original bands. Let the shape of the refactored data patch shape be (p, m, n), its relationship with the original data patch shape is as follows:

$$p = w \times h \quad m = (c + 2)/3 \quad n = (c + 2)/3. \tag{1}$$

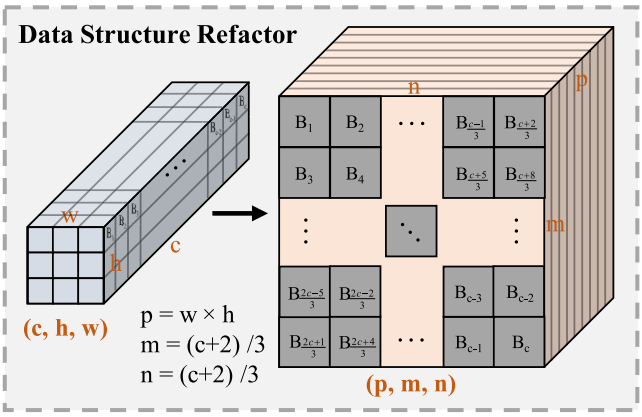


Fig. 1. Operation process details of the data structure refactor. The original data shape is (c, h, w), and the structure-refactored data shape is (p, m, n).

C. Design of Proposed SRDGnet

The overall structure of the proposed SRDGnet is shown in Fig. 2. It consists of five main components: SE-AEM, IFEH, domain shift reduction module (DSRM), IFEM-light, and a classifier. Among these, DSRM is newly introduced in SRDGnet, while the other four components retain the core concepts of their corresponding components in MDGTnet and have been lightweighted.

The key idea of the proposed SRDGnet is to change from separately computing domain environmental features and class features to focusing on the computation and reduction of domain shifts. Leveraging the principle that detailed features can enhance differences between domain environments while global features can weaken such differences, the structure-refactored data are input into SE-AEM to extract fine-grained local spectral features, and the original data are input into IFEH to extract global features. These features are both mapped to a lower dimensional feature space, respectively. Through DSRM, the two types of features are fused to achieve information complementarity and reduce domain shifts. Subsequently, IFEM-light is employed to extract class features, which are finally fed into the classifier for classification. The details of each component are introduced below.

1) SE-AEM: The design of SE-AEM aligns with the structure of the refactored data, serving as a refinement of the AEM in MDGTnet. The purpose is to reduce the number of parameters while improving the granularity of the extracted local spectral features, thereby providing more detailed information for subsequent domain shift calculations. When performing 2-D convolution on the original HSI data, global spectral features are extracted. However, when performing 2-D depth-wise separable convolution (DSC) on refactored data, fine-grained local spectral features can be extracted. Compared to feature extraction based on broad spectral segments (e.g., visible light and near-infrared), this method allows for more detailed and diverse spectral feature extraction by controlling the size of the convolution kernel, while also ensuring coherent features. The detailed structure of SE-AEM is shown in Fig. 3.

Specifically, the refactored data first undergoes feature extraction through a DSC-block and then passes through

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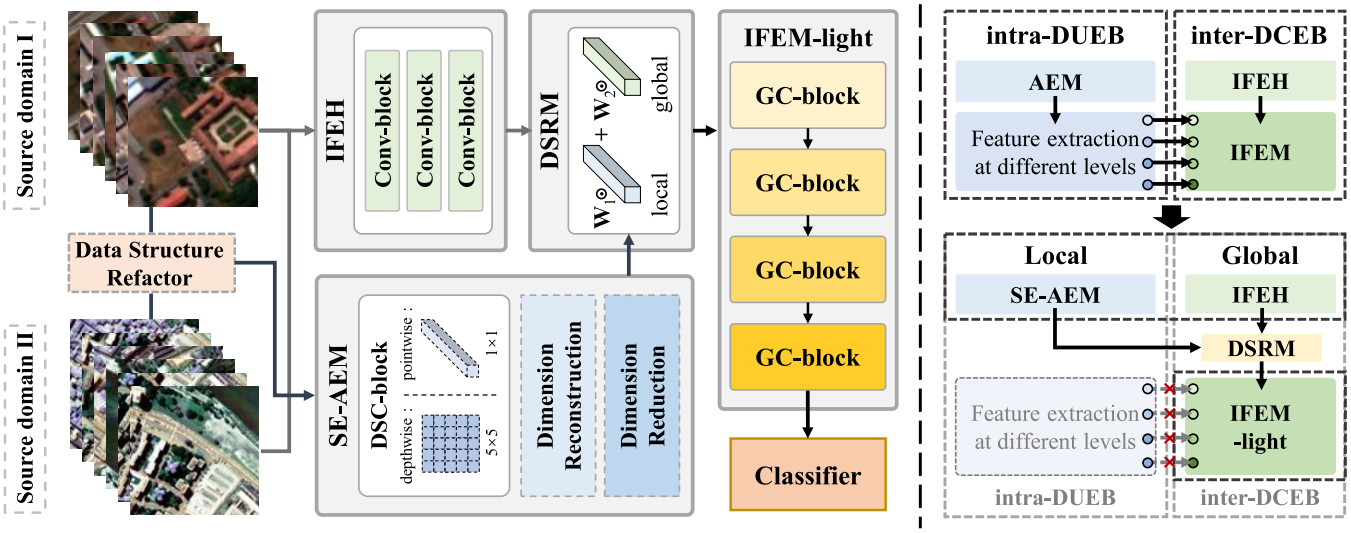


Fig. 2. Overall structure of the proposed SRDGnet. The key idea of the proposed SRDGnet is to change from separately computing domain environmental features and class features to focusing on the computation and reduction of domain shifts. It consists of five main components: SE-AEM, IFEH, DSRM, IFEM-light, and a classifier. The structure-refactored data are input into SE-AEM to extract fine-grained local spectral features, and the original data are input into IFEH to extract global features. Through DSRM, the two types of features are fused to achieve information complementarity and reduce domain shifts. Subsequently, IFEM-light is employed to extract class features, which are finally fed into the classifier for classification. On the right side are the differences between the proposed SRDGnet and MDGTnet in terms of the framework and modules.

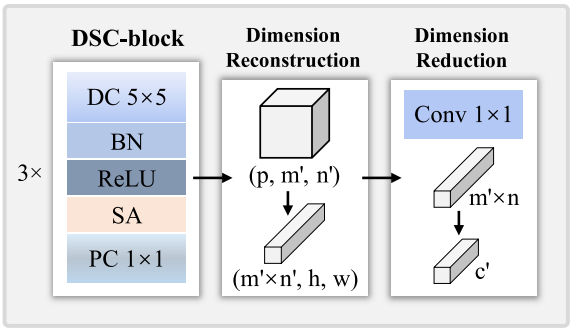


Fig. 3. Detailed structure of SE-AEM. The refactored data first undergoes feature extraction through a DSC-block and then passes through dimension reconstruction and reduction operations to output the final features.

TABLE I

PARAMETERS OF THE DC IN EACH CELL OF THE DSC-BLOCK AS WELL AS THE RECEPTIVE FIELD OF THE ACTION

RF	Unit 1	$RF_1 = 1+(k_1\|_{k=5}-1)\times s_1\|_{s=3}$=**13**
	Unit 2	$RF_2 = RF_1\|_{RF=13}+(k_2\|_{k=5}-1)\times s_1\|_{s=3}\times s_2\|_{s=1}$=**25**
	Unit 3	$RF_3 = RF_2\|_{RF=25}+(k_3\|_{k=5}-1)\times s_1\|_{s=3}\times s_2\|_{s=1}\times s_3\|_{s=1}$=**37**
RF_{spec}	Unit 1	$RF_{spec1} = 3\times RF_1\|_{RF=13}-2$=**37**
	Unit 2	$RF_{spec2} = 3\times RF_2\|_{RF=25}-2$=**73**
	Unit 3	$RF_{spec3} = 3\times RF_3\|_{RF=37}-2$=**109**

dimension reconstruction and reduction operations to output the final features. The DSC-block consists of three identical convolutional units, each including a 5 × 5 depth-wise convolution (DC), batch normalization (BN), a ReLU activation function, spatial attention (SA), and a 1 × 1 pointwise convolution (PC). The detailed parameters of the DC in each DSC-block unit, along with the receptive field and corresponding spectral receptive field changes, are presented in Table I. Here, *k* denotes the convolution kernel size, s represents the stride, RF refers to the receptive field of the convolution operation, and RF_{spec} represents the spectral receptive field corresponding to the original spectral dimension of the refactored data, indicating the range of spectral bands influenced by each convolution. The HSI data used in this article spans a spectral range of 380–1050 nm, mainly covering the visible light and near-infrared segments, with a spectral resolution of approximately 5 nm. Considering this, a 5 × 5 convolution kernel is chosen: the first unit affects spectral bands approximately covering half of each major spectral segment, the second unit extends to about the range of one major segment, and the third unit can cover the entire spectrum. This approach enables the extraction of fine-grained local spectral features without achieving excessive redundant information, while hierarchically integrating the spectral characteristics of major segments. It is worth noting that these parameters are not fixed and can be flexibly adjusted according to the specific characteristics (i.e., spectral range, number of bands, and spectral resolution) of the data.

In addition, to facilitate information interaction with the global features extracted from the original data, the features with spatial-spectral dimension swapping need to go through dimensional reconstruction. Precisely, the spatial dimension features are flattened, and the channel dimension is split and rearranged to match the original HSI patch pixel order, followed by a dimensional exchange. Finally, dimensionality reduction is performed to further condense the information.

2) IFEH: IFEH consists of three convolutional blocks (Conv-blocks), each comprising a 2-D convolution, BN, and ReLU (CBR). The convolution kernel size for the first CBR block is 3 × 3, while the kernel sizes for the subsequent two blocks are 1 × 1, all with a stride of 1.

3) DSRM: DSRM operates by dynamically reconciling the local spectral features extracted by SE-AEM with the global

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representations obtained by IFEH, facilitating a coordinated representation learning process. The local spectral features supplement the global features with detailed information. While detailed features delineate domain-specific characteristics, global representations contribute to cross-domain consistency. The proposed DSRM introduces a learning-driven feature alignment mechanism that adaptively calibrates complementary feature modalities, effectively mitigating domain shift by enhancing the model's sensitivity to informative variations and suppressing redundant domain-biased cues. Therefore, DSRM is designed with two learnable parameters as weighting coefficients for the two types of features. The functional process of DSRM can be expressed as follows:

$$\text{feat}_{\text{DSRM}} = \mathbf{W}_1 \odot \text{feat}_{\text{L}} + \mathbf{W}_2 \odot \text{feat}_{\text{G}} \tag{2}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ denote as two weighting coefficients, while feat_{L} and feat_{G} represent the local features and the global features from SE-AEM and IFEH, respectively. $\odot$ indicates the Hadamard product, which is element-wise multiplication.

4) IFEM-Light: In MDGTnet, IFEM is used to extract multilevel domain-invariant features, where feature alignment is achieved. The proposed SRDGnet retains its core functionality but introduces a lightweight variation, named IFEM-light. The IFEM-light is designed considering the following two aspects.

On the one hand, the features at each level in IFEM have high dimensions, requiring additional branches to refine the features further for similarity computation. However, the features fed into IFEM-light are dimensionally reduced, relatively low-dimensional, and sufficiently deep while preserving rich detail.

This makes the subsequent feature extraction do not require a large feature channel dimension to prevent the loss of information. Accordingly, the similarity computation can be directly performed using the multilevel deep features, and additional feature refinement branches are unnecessary, effectively reducing the network's parameter count. On the other hand, as network depth increases, features become increasingly "smooth," leading to the phenomenon of "blurring." To address the issue, group convolutions are introduced to enhance detail in the channel dimension, mitigating the blurring effect while reducing the network parameters.

In terms of specific module structure, IFEM-light includes four group convolutional blocks (GC-blocks), each comprising two group CBRs (GCBRs) and one CBR, along with residual connections to achieve feature fusion across different levels. Besides, channel attention (CA) is incorporated to refine the information. The detailed structure of the group conv-block is shown in Fig. 4, which also illustrates the changes in feature dimensions.

5) Classifier: To further reduce the number of network parameters, group convolution with the kernel size the same as the data patch is employed in the first two layers to decrease the channel dimensionality. Finally, a 1×1 convolution is applied to classification.

D. Construction Strategy of Sample Pairs

For contrastive learning, the number and diversity of sample pairs in the training data are crucial. To achieve higher diversity, the following sample pair construction strategy has been designed, with a detailed flowchart illustrated in Fig. 5.

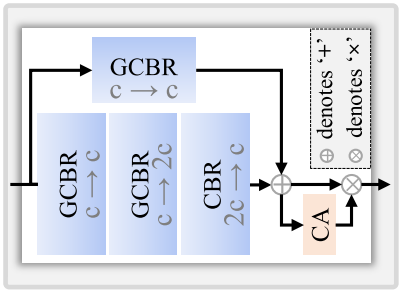


Fig. 4. Detailed structure of GC-block. IFEM-light includes four group convolutional blocks, each comprising two GCBRs and one CBR, along with residual connections to achieve feature fusion across different levels. CA is incorporated to refine the information further.

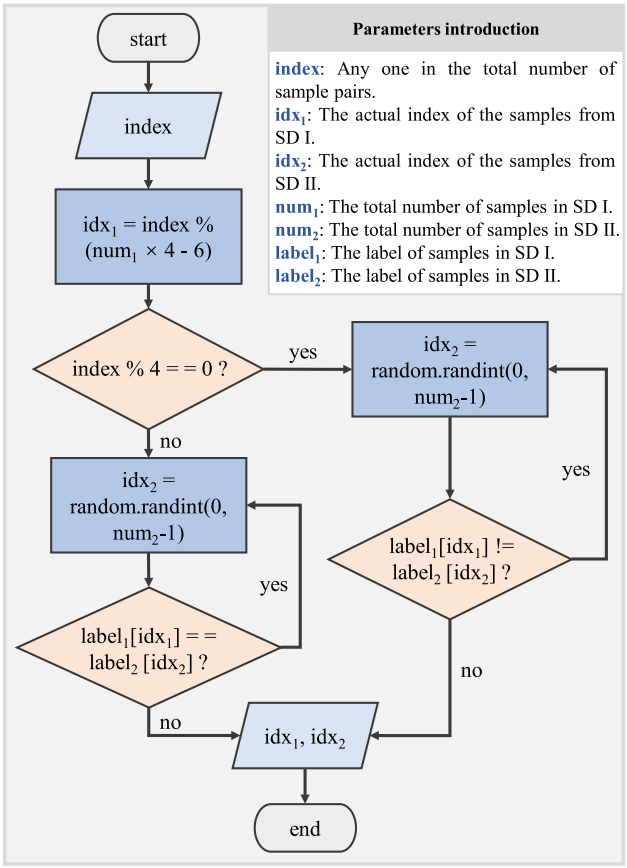


Fig. 5. Detailed operational flowchart of sample pair construction strategy.

Constructing sample pairs refers to the process of indexing samples from different SDs. Taking two SDs with four classes as an example, different class combinations result in four types of positive pairs and 12 types of negative pairs. Therefore, among all sample pairs, a 1:3 ratio of positive to negative pairs is maintained during construction. Specifically, the total number of sample pairs is first determined. Then, based on the rule that "when the index is a multiple of 4, samples of the same class are selected; otherwise, samples of different classes are chosen," the class of a sample in SD I is used as a reference to randomly index a corresponding sample from SD II. If the

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randomly selected sample does not meet the specified rule, the process is repeated until a valid sample is obtained. It is worth noting that, to further enhance the diversity of sample pairs, all samples in SD I are expanded fourfold through concatenation. This approach helps mitigate the decline in randomness caused by manually imposed constraints during indexing.

E. Loss Function

Similar to MDGTnet [48], the proposed method employs contrastive learning for model training and incorporates classification constraints, similarity constraints, and SDP constraints, with the latter two differing in specific details.

The classification loss is the binary cross-entropy (BCE) loss, denoted as l_{cls}. The expression is as follows:

$$\begin{aligned} l_{\text{cls}}(x,y) &= \frac{1}{C}\sum_{c=1}^{C}\frac{1}{N_c}\sum_{n_c=1}^{N_c} l_{n_c}(x,y) \\ l_{n_c}(x,y) &= -\left[w_c \cdot y_{n_c} \cdot \log x_{n_c} + \left(1-y_{n_c}\right)\cdot \log\left(1-x_{n_c}\right)\right] \end{aligned} \tag{3}$$

where C represents the number of classes. N_c denotes the number of samples in the cth class. w_c is the weight of the samples in the cth class. x_{n_c} and y_{n_c} represent the prediction and ground truth of the nth sample in the cth class, respectively. In general, multiclass classification tasks use cross-entropy (CE) as the classification loss for constraint. During training, CE can only explicitly constrain positive classes, while negative classes are implicitly constrained through the competition mechanism of the SoftMax. This belongs to passive constraint rather than independent optimization. The optimization of negative classes depends on the gradient propagation of positive classes, which may lead to insufficient suppression of negative classes during domain shift, especially when there are significant differences in class distributions between domains. Replacing CE with BCE as the classification loss allows both positive and negative classes to be considered simultaneously. BCE enables the classifier to achieve better performance. Moreover, from a testing perspective, CE loss assumes that the probabilities of different classes are mutually exclusive, meaning that test samples must belong to one of the predefined classes. However, when testing on an unseen TD image, a sample may belong to one of the training classes or none at all. BCE loss imposes independent binary constraints on each class, allowing classes to be treated separately. Considering the classifier's output probabilities, the constraints imposed by BCE loss produce more reasonable results and a more suitable choice for DG.

To encourage greater feature similarity for positive sample pairs and lower similarity for negative sample pairs, a cosine-based similarity loss is used for similarity computing to constrain the network. Unlike MDGTnet, to enhance data utilization, a batch-level pairwise traversal approach is employed to impose constraints on feature similarity. This loss function is denoted as l_{s}, and its expression is as follows:

$$\begin{aligned} l_{\text{s}}(s,k) &= -\frac{1}{B^2}\sum_{l=1}^{L}\sum_{b1=1}^{B}\sum_{b2=1}^{B} E_{l,b1,b2}(s,k) \\ E_{l,b1,b2}(s,k) &= k_{b1,b2}\cdot \log s_{b1,b2,l} + \left(1-k_{b1,b2}\right) \\ &\quad \cdot \log\left(1-s_{b1,b2,l}\right) \\ s_{b1,b2,l} &= \mathbf{X}_{1b1,l}\cdot \mathbf{X}_{2b2,l}, \mathbf{X}\in\mathbb{R}^{c\times h\times w}, s_{b1,b2,l}\in[0,1] \\ k_{b1,b2} &= \begin{cases} 1, & y_{1b1}=y_{2b2} \\ 0, & y_{1b1}\neq y_{2b2} \end{cases} \end{aligned} \tag{4}$$

where k represents the flag for positive or negative sample pairs, while y_{1b1} and y_{2b2} denote the labels of the $b1$th and $b2$th sample from two SDs, respectively. $\mathbf{X}_{1b1,l}$ and $\mathbf{X}_{1b2,l}$ are the features of the $b1$th and $b2$th sample at the lth layer of IFEM-light, and $s_{b1,b2,l}$ is their similarity value. B represents the batch size during training.

SDP loss is designed to constrain the change in the magnitude of similarity between higher and lower feature levels. When the input to the network is a positive sample pair, the similarity between the pair should increase as the feature hierarchy deepens. Conversely, when the input is a negative sample pair, the similarity should decrease while the dissimilarity increases as the feature hierarchy deepens. However, considering that the distances between different negative samples may naturally vary, the SDP loss is adjusted to maintain balance during the optimization process. Its adjusted expression is as follows:

$$\begin{aligned} l_{\text{SDP}}(s,k) &= \frac{1}{B^2}\sum_{l=1}^{L}\sum_{b1=1}^{B}\sum_{b2=1}^{B}\left(s_{b1,b2,l+1}-s_{b1,b2,l}-k_{b1,b2}\cdot w_l\right)^2 \\ s_{b1,b2,l} &= \mathbf{X}_{1b1,l}\cdot \mathbf{X}_{2b2,l}, \mathbf{X}\in\mathbb{R}^{c\times h\times w}, s_{b1,b2,l}\in[0,1] \\ k_{b1,b2} &= \begin{cases} 1, & y_{1b1}=y_{2b2} \\ 0, & y_{1b1}\neq y_{2b2} \end{cases} \end{aligned} \tag{5}$$

where w_l represents stands for the target threshold corresponding to the difference value in similarity from layer $l+1$ to layer l. The other variables have the same meaning as above.

The overall loss function expression is as follows:

$$l = \lambda_1\cdot l_{\text{cls}} + \lambda_2\cdot l_{\text{s}} + \lambda_3\cdot l_{\text{SDP}} \tag{6}$$

where λ_1, λ_2, and λ_3 are the weighting coefficients corresponding to the three losses, respectively.

IV. Experiments

In this section, the datasets and training settings are introduced first. The discussion on the generalization performance of the proposed model, the ablation study, and method comparison results are shown subsequently.

A. Datasets

Experiments have been conducted on four public HSI datasets: Houston 2013 (H13), Houston 2018 (H18), Pavia University (PU), and Pavia Center (PC).

1) *H13:* The Houston 2013 dataset was acquired using the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Contest. The image covers the University of Houston and its surrounding urban areas in the United States. The dataset has a spatial size of 349 × 1905 pixels with a spatial resolution of 2.5 m and includes 144 spectral bands spanning the wavelength range from 380 to 1050 nm.
2) *H18:* The Houston 2018 dataset, used in the 2018 IEEE GRSS Data Fusion Contest, was also acquired using the

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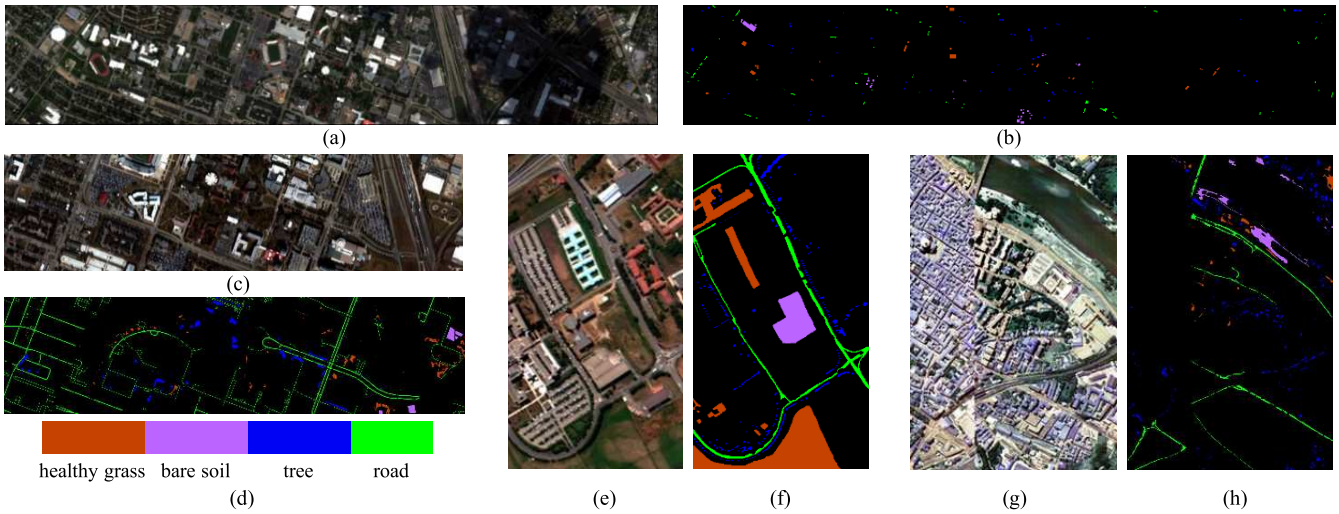


Fig. 6. False-color images and ground truth maps. (a) False-color image of H13 dataset. (b) Ground truth map of H13 dataset. (c) False-color image of H18 dataset. (d) Ground truth map of H18 dataset. (e) False-color image of PU dataset. (f) Ground truth map of PU dataset. (g) False-color image of PC dataset. (h) Ground truth map of PC dataset.

TABLE II
TOTAL NUMBER OF SAMPLES FOR EACH CLASS IN EACH DATASET

Class		Number of Samples			
No.	Name	H13	H18	PU	PC
1	Healthy grass	1251	9799	18649	3090
2	Bare soil	1242	4561	5029	6584
3	Tree	1244	13595	3064	7598
4	Road	1252	45866	6631	9248
Total		4989	73821	33373	26520

ITRES CASI-1500 sensor. The dataset has a spatial size of 601 × 2384 pixels with a spatial resolution of 1 m and includes 48 spectral bands covering the wavelength range from 380 to 1050 nm.

3) *PU:* The PU dataset was acquired by the ROSIS sensor during a flight campaign over Pavia in northern Italy. The spectral range of the imaging spans from 430 to 860 nm, with 103 spectral bands. The spatial size of the dataset is 610 × 340 pixels, and the spatial resolution is 1.3 m.
4) *PC:* The Pavia Center dataset was captured during the same flight campaign as the PU dataset, featuring a different scene. Its spectral range and spatial resolution are identical to those of the PU dataset, but with 102 spectral bands and a spatial size of 1096 × 715 pixels.

False-color images and ground truth maps for the four datasets are shown in Fig. 6. It should be noted that the ground truth maps only display the four common classes across all datasets, which are healthy grass (1), bare soil (2), tree (3), and road (4). The total number of samples for each class in each dataset is summarized in Table II.

B. Training Setting

1) Sample Selection and Band Processing: All datasets are divided into two groups. In the first group, H13 and H18 are treated as SDs, while PU and PC are regarded as TDs, written as G1 in the following description. In the second group, PU and PC are treated as SDs, while H13 and H18 are regarded as TDs, written as G2 in the subsequent discussion.

To ensure balance in the number of training samples across classes and considering the specific sample counts in each dataset, the sampling strategy is as follows: In the G1 experiments, all samples from H13 are used for training. For the H18 dataset, all samples from classes 1 and 2 are included, while 10 000 samples are randomly selected for each of classes 3 and 4. In the G2 experiments, for the PU dataset, all samples from classes 2 and 4 are included, while 5000 and 3000 samples are randomly selected for classes 1 and 3, respectively. For the PC dataset, all samples from classes 1 and 2 are used, while 6000 and 4000 samples are randomly selected for classes 3 and 4, respectively. Once the samples are selected, they remain fixed for all subsequent experiments without further changes.

To address the issue of different band numbers across the four datasets, the common spectral range shared by all datasets is selected. Regarding the PU dataset's 103 bands as a baseline, bands outside the common spectral range are removed, and the remaining bands are interpolated to 103. Specifically, for the H13 dataset, since its spectral range is broader and its resolution is slightly lower compared to the baseline, bands 5–107 are selected to approximately match the baseline. For the H18 dataset, the data are first upsampled by a factor of three using linear interpolation, and then processed in the same manner as H13. For the PC dataset, band 1 is duplicated at the beginning to ensure that the number of bands matches the baseline.

2) Hyperparameter Settings: In both sets of experiments, all hyperparameters are the same. The initial learning rates are both set to 0.02, and the batch sizes are both set to 1024. Under this condition, parameter sensitivity analyses about the weight coefficients of each loss and the input HSI patch size are conducted to evaluate the sensitivity of SRDGnet on the four TDs.

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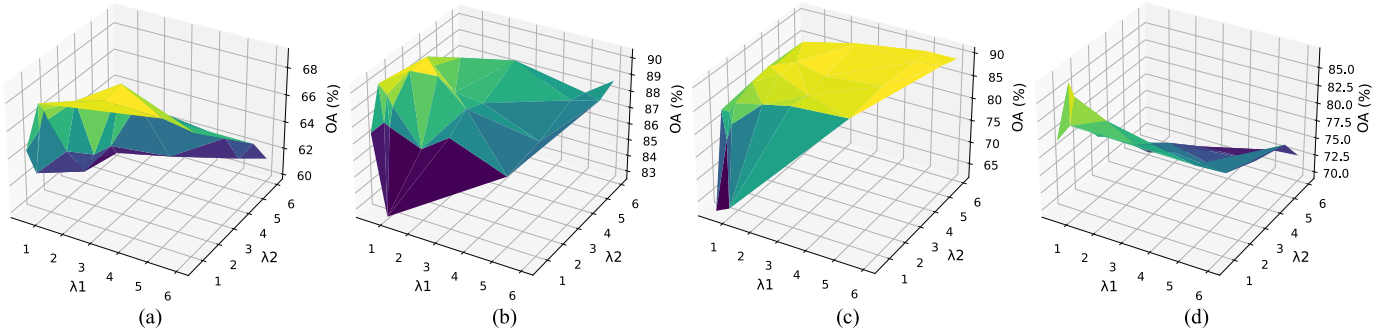


Fig. 7. Parameter tuning of λ_1 and λ_2 in the loss function. (a) H13+ H18-PU. (b) H13+ H18-PC. (c) PU + PC-H13. (d) PU + PC-H18.

TABLE III
CLASSIFICATION RESULTS (%) FOR DIFFERENT HSI PATCH SIZES

HSI patch size		1×1	3×3	5×5	7×7
H13+H18-PU	OA	50.91	**68.51**	64.48	62.95
	AA	60.34	**75.92**	74.11	72.04
	kappa	33.73	**54.45**	50.48	48.77
H13+H18-PC	OA	68.51	87.61	**89.60**	88.63
	AA	70.43	86.90	**89.75**	88.66
	kappa	57.93	83.01	**85.75**	84.43
PU+PC-H13	OA	67.18	89.35	**89.57**	88.83
	AA	67.11	89.35	**89.57**	88.83
	kappa	56.22	85.80	**86.09**	85.11
PU+PC-H18	OA	77.92	**81.75**	76.66	74.94
	AA	64.85	**87.52**	84.15	81.80
	kappa	61.29	**71.25**	64.47	62.24
Mean	OA	66.31	**81.81**	80.08	78.84
	AA	65.68	**84.92**	84.40	82.83
	kappa	52.29	**73.63**	71.70	70.14

Fixing λ_3 in (6), λ_1 and λ_2, regarded as adjustable hyperparameters, are selected from {6, 5, 3, 2, 1, 0.8, 0.5}, respectively. The experimental results on the four TDs are shown in Fig. 7. The optimal weights vary across different datasets, but overall, when λ_1 and λ_2 are set to {5, 1}, {3, 2}, {3, 1}, and {2, 1}, good classification performance is achieved on all four datasets. Through multiple experiments with different random seeds, {3, 1} is determined to be the best. Therefore, the values of λ_1, λ_2 and λ_3 in (5) are set to 3, 1, and 1, respectively.

Furthermore, to analyze the impact of the input HSI patch size on model performance, experiments have been conducted with patch sizes of 1×1, 3×3, 5×5, and 7×7, respectively. The experimental results on the four TDs are shown in Table III. Overall accuracy (OA), average accuracy (AA), and the kappa coefficient are used as evaluation metrics, and the same evaluation metrics are consistently applied in all subsequent classification experiments. The highest classification accuracy is highlighted in bold, and the second-best is underlined. This annotation rule is applied to all subsequent experimental results. On average, when the patch size is set to 3 × 3, the model achieves the best classification performance. Therefore, the input HSI patch size is set to 3 × 3.

3) Training Strategy: The proposed SRDGnet is trained on the PyTorch framework with a GeForce RTX3090. During training, the SDG optimizer is employed with a weight decay of 0.1 and momentum of 0.9. In G1, the learning rate is dynamically updated, decreasing by 5% every 120 iterations rather than training epochs. In G2, the learning rate is dynamically updated, decreasing by 15% every 80 iterations. An exponential moving average (EMA) training strategy is adopted, causing the model weights to oscillate around the actual optimal point during the last n epochs. By averaging the weights over these final n epochs, the model becomes more robust.

TABLE IV
MMD ACROSS DIFFERENT DOMAINS OF ORIGINAL DATA AND FEATURES EXTRACTED BY SRDGNET

Class	$mmd(SD_1, SD_2)$		$mmd(SD_1, TD)$		$mmd(SD_2, TD)$	
	origin	projection	origin	projection	origin	projection
1	1.0740	0.0234	1.2065	0.0424	1.2858	0.0232
2	0.6748	0.0359	1.5043	0.0233	0.5246	0.0238
3	1.0904	0.0115	1.2846	0.0265	0.8017	0.0158
4	1.0655	0.0089	1.5652	0.0285	1.1626	0.0171
Mean	0.9762	**0.0199**	1.3902	**0.0302**	0.9437	**0.0200**

C. Feature Extraction Capability of SRDGnet

The design of SRDGnet aims to extract domain-invariant discriminative features and improve the model's generalization ability. Taking the experiment where H13 and H18 datasets are employed as the SDs to train the model for classification on the TD PC dataset as an example, we analyze the separability of the original data and the features extracted by the model trained with SRDGnet. The results are visualized in Fig. 8. Different colors of the markers represent samples from different classes, with the numbers indicating the corresponding class index, and different shapes represent samples from different domains.

It is shown that the first line Fig. 8(a)–(c) represents the comparison of original data from each domain, revealing the phenomenon of "same object with different spectra, different objects with similar spectra" across domains. In contrast, the second line Fig. 8(d)–(f) shows the comparison of features extracted by SRDGnet, where the intra-class distances among samples from different domains are reduced, and inter-class distances are increased. It proves that the proposed SRDGnet has a strong ability to extract class discriminative features.

In addition, to quantify the distance of feature distributions, we compute the maximum mean discrepancies (MMDs) of

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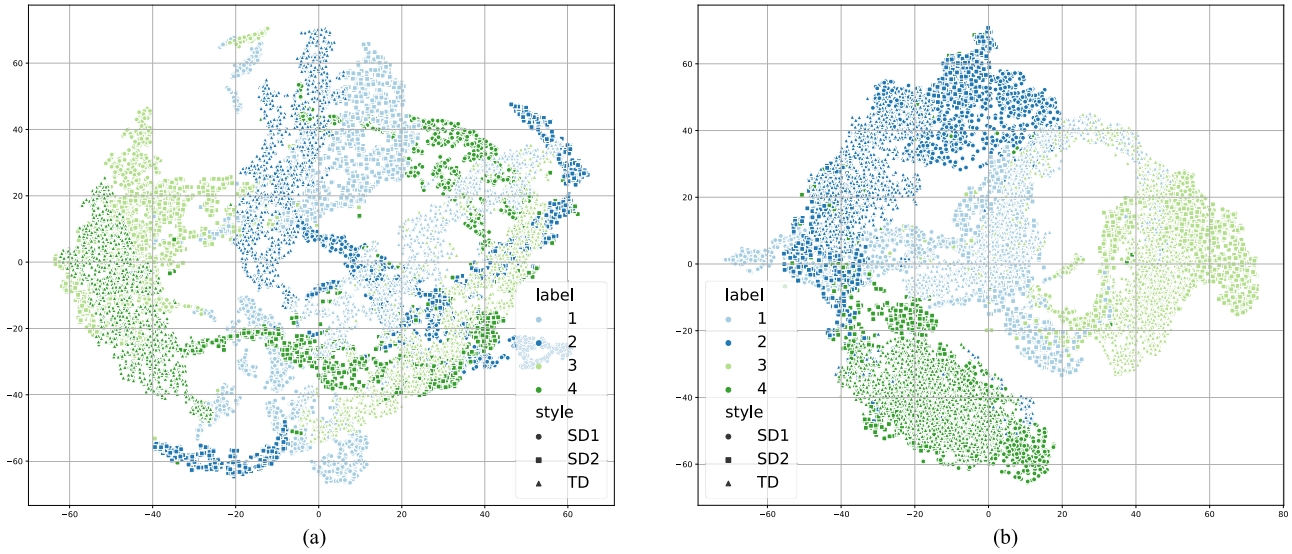


Fig. 8. Results of visualization of original data and features extracted by SRDGnet in different domains. (a) Original data from SD1, SD2, and TD. (b) Features extracted by SRDGnet of SD1, SD2, and TD.

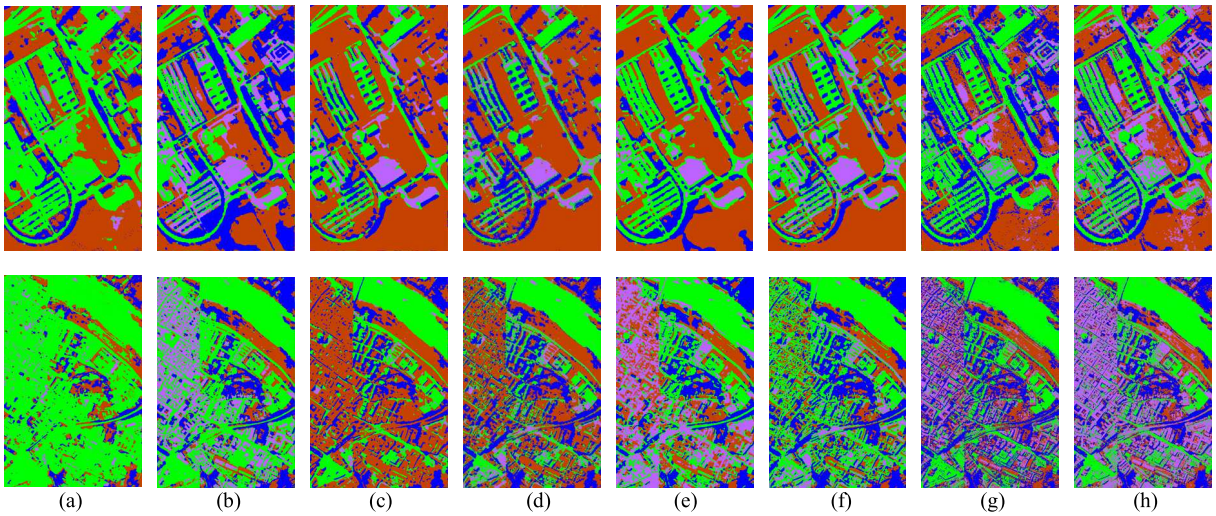


Fig. 9. Visualization results of the classification about different comparison methods on PU (first line) and PC dataset (second line). (a) TSTnet. (b) CLDA. (c) SCLUDA. (d) SDEnet. (e) LDGnet. (f) LLURnet. (g) MDGTnet. (h) Ours.

TABLE V

MODEL COMPONENT ABLATION CLASSIFICATION RESULTS (%) ON PU USING H13 AND H18 FOR TRAINING

	Model component				Class				OA	AA	kappa
	①	②	③	④	1	2	3	4			
a	×	×	×	×	47.42±3.71	47.79±2.98	94.80±3.18	95.40±1.29	61.36±2.18	71.35±1.69	46.84±2.18
b	×	√	√	√	58.72±2.54	49.75±3.21	98.68±0.40	95.63±1.44	68.37±1.00	75.69±0.70	54.28±1.05
c	√	×	√	√	58.92±3.96	48.79±3.34	98.84±0.27	96.23±1.70	68.47±2.27	75.69±1.24	**54.49±2.42**
d	√	√	×	√	52.60±5.69	46.28±3.19	96.90±3.49	96.49±1.36	64.44±3.12	73.07±1.82	49.98±3.29
e	√	√	√	×	57.86±4.02	51.39±3.95	98.70±0.94	95.26±2.00	68.06±1.81	75.80±0.86	54.04±1.86
f	√	√	√	√	58.64±2.86	49.57±2.44	98.85±0.44	96.64±1.31	**68.51±1.38**	**75.92±0.53**	54.45±1.43

features between two SDs as well as between SD and TD. Table IV presents the MMD values of the original data and SRDGnet-extracted features for each class across domains. The results show that the features extracted by SRDGnet are significantly more cohesive across all classes, which is beneficial for subsequent classification tasks.

D. Ablation Study

To evaluate the contributions of the key components and strategies of the proposed model, several ablation studies have been conducted by removing or replacing these elements. Specifically, six groups of experiments have been carried out, labeled from a to f. Among them, model f is trained by

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TABLE VI
MODEL COMPONENT ABLATION CLASSIFICATION RESULTS (%) ON PC USING H13 AND H18 FOR TRAINING

	Model component				Class				OA	AA	kappa
	①	②	③	④	1	2	3	4			
a	×	×	×	×	95.11±1.46	83.59±1.96	78.81±4.37	91.50±1.05	86.32±1.01	87.25±0.77	81.37±1.33
b	×	√	√	√	82.68±7.19	81.91±2.37	89.44±2.31	90.31±1.42	87.08±0.90	86.08±1.63	82.27±1.24
c	√	×	√	√	84.76±6.11	81.93±1.56	89.19±1.99	91.04±1.97	87.51±0.89	86.73±1.26	82.87±1.20
d	√	√	×	√	91.55±2.67	82.38±2.17	84.21±3.94	91.98±1.68	87.32±1.02	**87.53±0.75**	82.67±1.35
e	√	√	√	×	79.86±5.67	81.88±2.21	90.17±1.13	88.90±2.79	86.47±1.53	85.20±1.94	81.41±2.10
f	√	√	√	√	85.55±2.74	81.96±1.74	88.50±1.46	91.60±1.97	**87.61±0.90**	86.90±1.03	**83.01±1.23**

TABLE VII
MODEL COMPONENT ABLATION CLASSIFICATION RESULTS (%) ON H13 USING PU AND PC FOR TRAINING

	Model component				Class				OA	AA	kappa
	①	②	③	④	1	2	3	4			
a	×	×	×	×	76.80±3.00	95.15±1.22	92.15±2.38	96.95±0.32	**90.25±0.33**	**90.26±0.33**	**87.00±0.44**
b	×	√	√	√	73.55±7.98	90.41±3.09	93.04±3.46	98.21±0.48	88.79±1.62	88.80±1.61	85.06±2.16
c	√	×	√	√	65.40±11.41	98.17±0.57	97.11±1.32	94.84±1.25	88.85±2.74	88.88±2.73	85.14±3.65
d	√	√	×	√	79.66±3.14	91.33±2.87	90.40±3.91	97.91±0.49	89.83±0.97	89.83±0.97	86.43±1.29
e	√	√	√	×	69.88±6.78	94.08±1.53	94.54±2.49	97.68±0.28	89.03±1.36	89.04±1.36	85.37±1.82
f	√	√	√	√	77.76±5.57	89.62±2.28	91.66±3.90	98.38±0.16	89.35±1.22	89.35±1.22	85.80±1.63

TABLE VIII
MODEL COMPONENT ABLATION CLASSIFICATION RESULTS (%) ON H18 USING PU AND PC FOR TRAINING

	Model component				Class				OA	AA	kappa
	①	②	③	④	1	2	3	4			
a	×	×	×	×	99.36±0.41	87.28±14.71	72.65±5.75	68.75±2.84	74.67±2.19	82.01±3.66	62.07±2.79
b	×	√	√	√	93.44±6.00	97.40±0.85	81.30±5.94	75.47±2.49	80.27±1.69	86.90±1.08	69.34±2.26
c	√	×	√	√	91.35±7.68	99.73±0.13	87.14±4.16	58.61±3.99	70.73±2.64	84.21±1.85	58.35±3.00
d	√	√	×	√	98.91±1.01	93.63±9.22	70.55±8.98	74.63±2.26	78.26±2.87	84.43±3.32	66.45±3.99
e	√	√	√	×	97.25±3.24	97.87±0.95	79.16±5.44	72.83±2.62	78.77±0.94	86.78±0.90	67.48±1.08
f	√	√	√	√	96.64±3.66	96.39±1.09	79.16±4.92	77.90±2.16	**81.75±1.28**	**87.52±0.78**	**71.25±1.71**

TABLE IX
MMD ACROSS DIFFERENT DOMAINS OF MODEL d AND MODEL f

Class	$mmd(SD_1, SD_2)$		$mmd(SD_1, TD)$		$mmd(SD_2, TD)$	
	model d	model f	model d	model f	model d	model f
1	0.0258	0.0234	0.0270	0.0424	0.0284	0.0232
2	0.0498	0.0359	0.0427	0.0233	0.0538	0.0238
3	0.0045	0.0115	0.0285	0.0265	0.0325	0.0158
4	0.0079	0.0089	0.0400	0.0285	0.0326	0.0171
Mean	0.0220	**0.0199**	0.0346	**0.0302**	0.0368	**0.0200**

the proposed method. The variables in the ablation studies are denoted as ① to ④. Under the framework of SRDGnet, ① indicates whether SE-AEM is used, ② indicates whether DSRM is employed, ③ represents whether IFEM and the classifier are designed in a lightweight manner, and ④ denotes whether the improved similarity and SDP constraints are applied instead of those from MDGTnet. "Yes" is marked as "√" and "No" is marked as "×." The classification results of these six groups of experiments on different TDs are presented in Tables V–VIII, with the specific variations of the four variables detailed in the tables. All experiments have been conducted with ten training runs using different random seeds, and the mean and standard deviation of the test results are reported.

It can be seen that model a, which lacks all the improved components and constraints, demonstrates the poorest overall performance. Although it achieves a relatively good classification result on the TD H13 dataset, its classification accuracy on the TD PU dataset is very low, while its accuracy on H18 dataset is not good enough, with large standard deviations for the second and third classes, indicating unstable performance. This proves that the performance of model a is heavily dependent on the TD being tested, indicating its limited generalization ability. When one improved component or constraint is removed from the proposed method, resulting in model b through model e, their overall performance improves significantly compared to model a, but shows a noticeable decline compared to model f. This demonstrates the effectiveness of each improved component and constraint. Among these four models, model c, which excludes DSRM, exhibits the most significant performance drop compared to model f. Specifically, its classification ac-curacy for the first class on the TD H13 dataset decreases substantially, with a large standard deviation, and its classification performance on H18 dataset also deteriorates markedly.

To further verify the effectiveness of IFEM-light in mitigating the feature blurring problem caused by increasing network depth, evaluations have been conducted not only in terms of classification accuracy but also by calculating the MMDs of

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TABLE X

TOTAL NUMBER OF PARAMETERS IN THE NETWORK WITH DIFFERENT FEATURE DIMENSIONS AND THE CLASSIFICATION RESULTS (%) ON DIFFERENT TDS

Feature Dimension	64			128			256			512		
Params	0.17M			0.45M			1.37M			4.58M		
Metrics	OA	AA	kappa	OA	AA	kappa	OA	AA	kappa	OA	AA	kappa
H13+H18-PU	67.35	75.10	53.05	68.51	**75.92**	54.45	**68.68**	75.45	**54.52**	67.03	74.98	52.89
H13+H18-PC	85.56	85.20	80.24	87.61	86.90	83.01	87.01	86.70	82.22	**87.70**	**87.47**	**83.15**
PU+PC-H13	86.60	86.62	82.14	89.35	89.35	85.80	**89.79**	**89.79**	**86.39**	89.64	89.64	86.19
PU+PC-H18	81.10	86.50	70.29	**81.75**	**87.52**	**71.25**	80.90	87.11	70.10	80.42	86.10	69.29
Mean	80.15	83.36	71.43	**81.81**	**84.92**	**73.63**	81.60	84.76	73.31	81.20	84.55	72.88

TABLE XI

CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON FOUR TDS

SD-TD	Class	TSTnet	CLDA	SCLUDA	SDEnet	LDGnet	LLURnet	MDGTnet	SRDGnet
H13+H18-PU	1	76.74±20.40	61.45±7.42	95.04±3.98	**99.10±1.16**	95.93±5.48	96.35±2.10	63.92±11.99	58.64±2.86
	2	15.97±6.32	25.29±8.63	16.70±3.80	9.00±7.15	4.50±5.64	6.80±4.64	40.66±7.83	**49.57±2.44**
	3	67.22±25.44	**98.99±0.48**	85.31±10.8	53.07±20.33	46.99±28.49	24.23±7.52	92.77±7.35	98.85±0.44
	4	86.86±13.93	**97.94±1.43**	90.01±5.10	53.22±15.16	82.92±7.28	89.98±6.00	95.00±5.93	96.64±1.31
	OA	68.72±10.38	66.70±4.18	**81.34±1.97**	72.18±3.05	75.07±2.08	74.96±1.53	69.24±6.12	68.51±1.38
	AA	61.70±5.71	70.92±2.58	71.76±2.24	53.60±5.92	57.58±6.11	54.34±2.76	73.09±1.98	**75.92±0.53**
	kappa	50.72±11.00	50.86±4.68	**67.50±2.82**	48.58±6.51	54.43±4.00	54.10±2.73	55.00±6.32	54.45±1.43
H13+H18-PC	1	87.00±20.38	56.60±29.95	**93.40±8.83**	90.36±5.36	62.99±28.51	85.45±2.76	92.08±10.20	85.55±2.74
	2	47.43±20.46	57.44±27.88	29.28±11.53	27.20±16.35	20.57±20.02	21.98±14.67	70.67±7.91	**81.96±1.74**
	3	56.75±35.72	76.92±39.08	80.33±11.67	78.50±12.66	82.48±28.10	55.48±11.25	72.71±15.51	**88.50±1.46**
	4	92.44±10.03	**99.27±0.68**	93.74±2.92	40.62±15.92	68.31±20.58	82.24±6.97	90.31±7.37	91.60±1.97
	OA	70.41±9.77	77.51±11.33	73.86±3.93	53.94±5.39	59.90±5.49	59.98±4.51	80.60±4.65	**87.61±0.90**
	AA	70.90±8.52	72.56±11.17	74.19±3.90	59.17±3.84	58.59±4.49	61.29±3.96	81.44±3.82	**86.90±1.03**
	kappa	59.86±12.43	69.61±14.74	65.34±4.98	40.47±6.30	46.82±6.33	46.52±5.55	73.87±5.98	**83.01±1.23**
PU+PC-H13	1	72.37±15.62	43.23±32.78	83.43±3.25	**88.70±6.22**	68.61±22.44	72.44±14.28	70.80±10.86	77.76±5.57
	2	73.36±24.65	83.68±23.41	90.15±29.49	**99.71±0.46**	67.10±41.76	99.25±1.93	90.35±4.47	89.62±2.28
	3	38.44±36.24	80.87±18.54	65.46±31.37	7.54±7.20	0.79±2.36	45.96±25.46	87.57±7.39	**91.66±3.90**
	4	85.91±23.27	59.27±39.52	57.50±31.10	25.50±22.90	46.79±36.56	40.41±14.74	98.03±0.82	**98.38±0.16**
	OA	67.56±6.20	66.71±9.74	74.12±14.91	55.34±6.06	45.85±7.82	64.72±7.37	86.68±1.94	**89.35±1.22**
	AA	67.52±6.20	66.76±9.74	74.14±14.91	55.36±6.04	45.82±7.80	64.77±7.38	86.69±1.94	**89.35±1.22**
	kappa	56.73±8.27	55.62±12.99	65.49±19.88	40.46±8.07	27.80±10.41	52.97±9.83	82.24±2.59	**85.80±1.63**
PU+PC-H18	1	94.91±8.72	63.85±41.64	**99.88±0.30**	35.39±24.23	29.67±24.16	52.31±19.89	82.12±10.69	96.64±3.66
	2	72.58±19.04	82.30±16.08	78.35±22.00	96.54±5.99	32.87±32.73	66.23±19.72	**99.50±0.49**	96.39±1.09
	3	60.30±24.08	82.07±11.29	60.75±18.17	9.12±5.53	10.61±20.92	29.83±14.79	**89.44±5.29**	79.16±4.92
	4	65.37±23.12	46.88±17.36	46.38±16.36	51.69±15.49	**82.14±15.83**	56.02±21.52	61.36±5.08	77.90±2.16
	OA	68.80±12.16	57.79±14.78	58.09±10.26	44.43±9.56	58.97±8.95	51.32±12.63	71.62±3.71	**81.75±1.28**
	AA	73.29±5.98	68.77±14.61	71.34±6.20	48.19±6.44	38.82±11.07	51.10±7.19	83.10±2.77	**87.52±0.78**
	kappa	54.93±12.51	43.46±16.02	44.21±9.48	25.52±7.53	33.16±9.93	32.24±10.48	59.00±4.20	**71.25±1.71**

TABLE XII

COMPARISON OF THE PROPOSED SRDGNET AND MDGTNET IN TERMS OF PARAMETER COUNT, RUNTIME, AND COMPUTATIONAL COST

Method	MDGTnet	SRDGnet
training time	564.3s	**294.4s**
testing time	4s	**3s**
FLOPs	0.22G	**0.0074G**
Params	14.84M	**0.4543M**

features between the two SDs and between the SD and TD before and after ablation. The experiment where the H13 and H18 datasets have been used as SDs to train the model for classification on the TD PC dataset is taken as an example. The MMD between different domains for model d and model f are presented in Table IX. The results indicate that the intra-class distances of model d are higher than those of model f, and the differences in MMD values among various classes are larger than those of model f. It suggests that model d lacks clear optimization directions to some extent and tends to favor certain classes, leading to more feature blurring compared to model f.

What is more, to explore the impact of feature dimensions and model parameters on the generalization performance, we set the input and output feature dimensions of IFEM-light to 64, 128, 256, and 512, respectively. We have calculated the model parameters and recorded the average classification performance of each model across different TDs, as shown in Table X.

It can be observed that when the feature dimension is set to 128, the model achieves the best average classification performance. When the feature dimension is increased to 256, the average classification accuracy slightly decreases, although performance on certain TDs shows minor improvements. However, this also results in a noticeable increase in the number of network parameters. Determining whether

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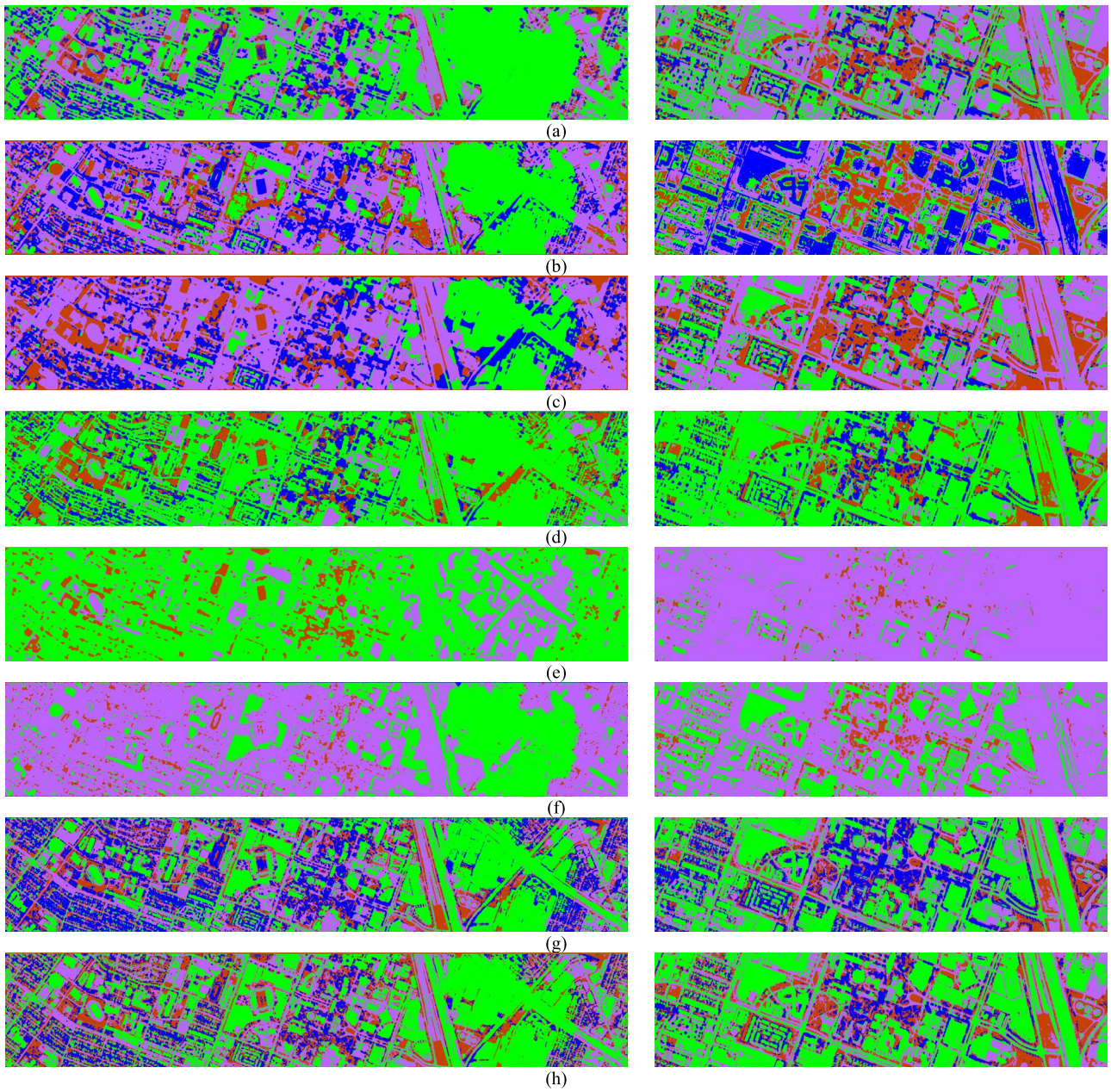


Fig. 10. Visualization results of the classification about different comparison methods on H13 (left column) and H18 dataset (right column). (a) TSTnet, (b) CLDA. (c) SCLUDA. (d) SDEnet. (e) LDGnet. (f) LLURnet. (g) MDGTnet. (h) Ours.

to allocate additional computational resources for marginal improvements in classification performance on certain datasets remains an important consideration.

After balancing all aspects, 128 is chosen as the feature dimension for the proposed SRDGnet. It is worth noting that these experiments have been conducted under the specific data volume of the current datasets. When using other datasets with larger or smaller data volumes, the feature dimensions should be adjusted accordingly to achieve optimal results.

E. Method Comparison

To validate the effectiveness, the proposed SRDGnet is compared with seven state-of-the-art methods: TSTnet [27], CLDA [32], SCLUDA [26], SDEnet [42], LDGnet [43], LLURnet [44], and MDGTnet [48]. Each method is evaluated through ten experiments with different random seeds, and the mean and standard deviation of the results are calculated. Among these methods, TSTnet, CLDA, and SCLUDA are DA methods; SDEnet, LDGnet, and LLURnet are single-source DG methods; and MDGTnet represents a multisource DG method. All comparison methods train and test on the same datasets, with the codes sourced from the respective papers.

1) Quantitative Comparison: The classification results of different methods on the four TDs are presented in Table XI. It can be seen that the proposed SRDGnet achieves the highest accuracy and consistency on three datasets, demonstrating an apparent advantage. Although the OA on the TD PU dataset is not the highest, the accuracy across all classes remains sat-

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TABLE XIII
CLASSIFICATION RESULTS (%) ON PU OF DIFFERENT SINGLE-SOURCE DG METHODS TRAINED BY SINGLE SD

SD-TD	Class	SDEnet	LDGnet	LLURnet
H13-PU	1	76.18±18.24	91.71±11.96	96.47±1.94
	2	55.59±21.50	6.61±4.51	23.81±9.30
	3	49.41±26.21	62.18±28.20	43.42±19.70
	4	24.99±16.68	56.62±19.72	34.68±2.12
	OA	60.45±7.84	69.20±6.09	68.37±2.35
	AA	51.54±8.74	54.28±6.70	49.59±6.25
	kappa	37.29±9.76	45.86±7.97	43.66±5.21
H18-PU	1	98.82±1.65	96.09±3.78	98.48±1.14
	2	4.63±2.62	2.20±1.75	8.32±5.49
	3	46.21±21.75	66.66±16.07	17.70±7.38
	4	74.69±6.01	58.75±17.03	69.29±7.34
	OA	**75.00±1.72**	**71.82±3.50**	**71.68±2.10**
	AA	56.09±5.26	55.93±5.95	48.45±4.12
	kappa	**53.62±3.97**	47.82±6.69	46.47±4.72
PC-PU	1	41.61±4.47	53.99±9.10	54.03±5.76
	2	93.84±2.40	91.62±6.65	86.59±4.13
	3	69.35±5.23	51.93±27.89	77.10±3.29
	4	96.18±1.72	81.84±17.45	93.06±2.08
	OA	62.87±1.98	65.00±5.95	68.82±2.61
	AA	**75.24±1.20**	**69.85±10.41**	**77.69±1.02**
	kappa	49.46±1.92	**49.95±8.66**	**55.81±2.77**

TABLE XIV
CLASSIFICATION RESULTS (%) ON PC OF DIFFERENT SINGLE-SOURCE DG METHODS TRAINED BY SINGLE SD

SD-TD	Class	SDEnet	LDGnet	LLURnet
H13-PC	1	89.97±9.05	47.71±30.57	85.25±10.92
	2	90.10±9.55	15.56±13.70	78.31±11.84
	3	63.62±27.39	90.53±11.67	70.51±18.20
	4	23.60±11.85	49.86±31.80	31.83±8.14
	OA	59.31±8.67	52.75±9.41	60.68±4.55
	AA	66.82±6.23	50.92±8.85	66.48±3.29
	kappa	47.36±10.21	36.57±11.87	48.49±5.19
H18-PC	1	84.89±10.98	78.23±15.93	89.33±3.55
	2	11.84±6.07	13.26±4.68	15.94±11.96
	3	83.96±12.25	85.37±11.09	38.44±11.38
	4	60.67±17.31	51.69±20.77	61.08±4.70
	OA	58.04±6.19	54.89±7.17	46.68±4.11
	AA	60.34±4.60	57.14±5.61	51.20±3.89
	kappa	44.55±7.20	41.47±8.62	31.97±4.70
PU-PC	1	7.58±2.91	12.44±8.61	4.30±1.67
	2	88.10±6.95	77.40±8.40	90.64±4.04
	3	99.99±0.02	99.52±1.04	100±0.00
	4	99.66±0.39	97.53±3.12	99.88±0.08
	OA	**86.16±1.79**	**83.19±2.58**	**86.48±1.09**
	AA	**73.83±1.90**	**71.72±3.10**	**73.70±1.22**
	kappa	**80.22±2.61**	**76.01±3.72**	**80.69±1.59**

TABLE XV
CLASSIFICATION RESULTS (%) ON H13 OF DIFFERENT SINGLE-SOURCE DG METHODS TRAINED BY SINGLE SD

SD-TD	Class	SDEnet	LDGnet	LLURnet
PU-H13	1	6.31±7.20	55.12±39.31	0.97±1.94
	2	80.93±20.24	77.36±34.45	92.47±15.05
	3	86.18±18.52	28.75±26.11	49.84±23.97
	4	42.92±37.03	36.62±34.82	44.90±33.27
	OA	53.99±4.93	49.44±9.81	46.96±8.65
	AA	54.08±4.91	49.46±9.78	47.05±8.62
	kappa	38.70±6.56	32.61±13.05	29.34±11.50
PC-H13	1	91.53±5.02	58.89±32.87	75.34±14.93
	2	99.72±0.56	99.22±2.29	99.40±1.79
	3	11.41±8.34	0.54±1.61	22.84±21.18
	4	18.86±16.73	5.98±12.99	25.13±19.53
	OA	55.35±4.41	41.11±9.39	55.64±9.80
	AA	55.38±4.40	41.16±9.37	55.68±9.79
	kappa	40.48±5.88	21.53±12.48	40.87±13.06
H18-H13	1	64.80±19.18	75.68±12.66	83.51±2.64
	2	96.41±1.22	93.43±4.63	99.14±0.96
	3	94.13±4.54	88.69±7.01	76.87±3.61
	4	98.13±1.28	97.78±2.64	96.54±1.41
	OA	**88.35±3.94**	**88.89±2.90**	**89.02±0.54**
	AA	**88.37±3.92**	**88.90±2.90**	**89.02±0.54**
	kappa	**84.46±5.25**	**85.19±3.87**	**85.36±0.72**

TABLE XVI
CLASSIFICATION RESULTS (%) ON H18 OF DIFFERENT SINGLE-SOURCE DG METHODS TRAINED BY SINGLE SD

SD-TD	Class	SDEnet	LDGnet	LLURnet
PU-H18	1	9.99±10.06	13.81±28.28	6.47±4.41
	2	50.24±19.96	39.65±36.39	67.19±28.99
	3	68.18±14.80	61.00±15.55	47.48±18.51
	4	55.02±35.28	80.78±20.82	66.86±22.00
	OA	51.17±18.73	65.72±10.72	55.29±13.17
	AA	45.86±5.42	48.81±8.54	47.00±7.21
	kappa	33.44±14.45	44.93±10.91	35.59±11.69
PC-H18	1	54.30±24.77	23.68±29.79	41.37±8.73
	2	87.31±21.50	56.35±40.07	87.91±10.52
	3	14.59±10.67	1.80±3.32	18.87±10.88
	4	52.31±18.27	57.95±25.45	54.76±13.78
	OA	48.07±11.91	42.95±16.50	48.40±6.85
	AA	52.13±6.15	34.95±8.06	50.73±3.52
	kappa	30.12±9.74	21.22±12.42	32.04±8.92
H13-H18	1	92.87±6.79	79.26±19.25	99.55±0.42
	2	62.15±16.88	55.72±21.32	87.27±12.07
	3	90.15±4.04	79.94±20.80	78.85±7.69
	4	94.92±0.77	90.62±5.48	93.17±2.09
	OA	**91.77±1.27**	**84.94±2.97**	**91.02±2.16**
	AA	**85.02±4.31**	**76.38±4.87**	**89.71±3.09**
	kappa	**85.32±2.37**	**73.94±4.84**	**84.48±3.56**

isfactory. In contrast, most other methods, except MDGTnet, perform poorly on the second class, with the three single-source DG methods being the most severely affected. The proposed SRDGnet achieves the highest AA on all TDs, which also reflects its superior performance across various classes. Moreover, SRDGnet maintains consistently good performance across all TDs, whereas the classification performance of other DA and single-source DG methods varies significantly among the different TDs. Besides, SRDGnet exhibits relatively small variances compared to other methods, indicating strong model stability. Notably, compared with the multisource DG method MDGTnet, the proposed SRDGnet achieves improvements in accuracy, consistency, and stability, while requiring fewer network parameters, lower computational resources, and shorter training and testing times. The specific values for parameters, time, and other metrics are presented in Table XII.

Consequently, the experimental results demonstrate that the proposed method can obtain a model with superior generalization ability, while the other models trained by comparison methods display weaker generalization and are more heavily affected by the dataset.

In addition, to further illustrate the necessity of multisource DG, a series of experiments using the three single-source DG methods to train with a single SD have been conducted. The classification results of different methods on four TDs are shown in Tables XIII–XVI. Combining with the experimental

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results in Table XI, where multi-SD training is used, it can be observed that for the same TD, the classification results of the multi-SD trained model are roughly a compromise of the performance of models trained with each single SD. The classification performance of the single-source DG model is directly related to the selection of the SD data used in training. In practical applications, which SD will yield the best results for training cannot be predicted, but we can choose as many available SDs as possible to improve the average performance of the model, to handle various unknown TDs.

2) Visual Comparison: The visualization results of the classification about different comparison methods are shown in Figs. 9 and 10. In the result images for different TDs, it can be seen that the proposed SRDGnet achieves a performance with higher consistency and less noise.

V. Conclusion

In this article, the SRDGnet is proposed for HSI cross-domain classification, designed with a lightweight network in terms of focusing on reducing domain shift to mitigate the overfitting phenomenon. In addition, the data representation of the HSI patch and the modules included in SRDGnet are also improved to extract finer features and reduce the number of network parameters. Moreover, batch-wise constraints are employed to increase the sufficiency of data utilization. Finally, experimental results on four different TDs reflect the model's good generalization ability and stability, demonstrating the effectiveness of the proposed method. However, under conditions of limited data diversity and quantity, the performance improvement is also limited. In the future, we will consider using more abundant data to further enhance the generalization ability of the model.

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